Vadodara institute of engineering
kotambi
BE sem 6
Mad important qbank

Q1. Android Fundamentals What is Android? Explain its origin, key features, and the role of the Open Handset Alliance. How does Android differ from other mobile operating systems in terms of openness, licensing, and architecture?

Q2. Application Components Describe the four core Android Application Components — Activity, Service, Broadcast Receiver, and Content Provider. Explain the purpose of each component and give one real-world use case for each.

Q3. The Android Manifest File What is the AndroidManifest.xml file and why is it essential to every Android application? List and explain at least six declarations that must be made in the manifest, including components, permissions, and hardware features.

Q4. Development Environment & First App Explain the steps involved in setting up the Android development environment — from downloading Android Studio to running your first application. What is the role of the AVD Manager, Gradle, and the Logcat window during development?

Q5. Activity & Fragment Lifecycle Explain the complete lifecycle of an Android Activity with all callback methods (onCreate to onDestroy) and the scenarios that trigger each. How do Fragments differ from Activities, and what is the Fragment lifecycle?

Q6. Intents — Explicit & Implicit What is an Intent in Android? Differentiate between Explicit and Implicit Intents with examples. How is data passed between Activities using Intent extras? What is an Intent Filter and how does Android resolve which component handles an Implicit Intent?

Q7. Built-in Application Invocation How does the Intent object allow you to invoke Android's built-in applications such as the dialer, browser, camera, and maps? Explain the role of action strings (e.g., ACTION_DIAL, ACTION_VIEW) and URI data in triggering these built-in apps.

Q8. Views, ViewGroups & Layouts What is the difference between a View and a ViewGroup in Android? Explain the purpose and XML attributes of LinearLayout, RelativeLayout, ConstraintLayout, and GridLayout. Which layout is recommended for complex, deeply nested UIs and why?

Q9. AdapterView, AutoCompleteTextView & Screen Orientation What is an AdapterView? Explain how the Adapter pattern works with ListView and Spinner. What is AutoCompleteTextView and how does it provide suggestions? How do you handle Screen Orientation changes in Android to preserve the UI state?

Q10. UI Event Handling & Menus Explain event-driven programming in Android. Describe the event handling methods: onClick(), onTouch(), onKey(), and onFocusChange(). Differentiate between the three types of menus — Options Menu, Context Menu, and Popup Menu — and explain how each is created and handled.

Q11. Data Storage Options What are the different data storage options available in Android? Compare Internal Storage, External Storage, Shared Preferences, SQLite Database, and Network/Cloud Storage in terms of data scope, persistence, capacity, and appropriate use cases.

Q12. Internal & External Storage Explain how to read and write files using Android's Internal Storage. How does External Storage differ from Internal Storage in terms of accessibility and permissions? What checks must be performed before accessing external storage?

Q13. SQLite Database Explain how SQLite is used in Android for structured data storage. What is the role of SQLiteOpenHelper? Describe all four CRUD operations — Insert, Read/Query, Update, and Delete — including the key classes and methods involved (ContentValues, Cursor, rawQuery, execSQL).

Q14. Content Providers What is a Content Provider in Android? How does it enable data sharing between applications? Explain the role of a ContentResolver, URI scheme, and the CRUD methods (query, insert, update, delete) exposed by a Content Provider. Give an example of a built-in Content Provider.

Q15. Google Maps & Location Services How do you integrate Google Maps into an Android application? Explain the steps: obtaining an API key, adding the Maps SDK dependency, configuring the manifest, and displaying a map with a marker. What permissions are required for location access?

Q16. Geocoding & Reverse Geocoding What is Geocoding and Reverse Geocoding? How does the Android Geocoder class support both operations? Explain the difference between converting an address to coordinates and coordinates to an address, and describe a practical scenario where each is used.

Q17. Graphics, Drawables & Animation Explain the use of Drawable objects in Android UI. What is the difference between a BitmapDrawable and a ShapeDrawable? What is Hardware Acceleration in Android and how does it improve rendering performance? Describe the two main animation systems in Android: View Animation and Property Animation (Animator framework).

Q18. Audio, Video & Camera Explain the role of the MediaPlayer class in Android for playing audio and video. What is a SoundPool and how does it differ from MediaPlayer in terms of use case and performance? How is the Camera API used to capture images and record video in Android?

Q19. Recording Audio & Video How does the MediaRecorder class work in Android? Describe the sequence of method calls required to record audio and then video. What permissions and manifest entries are required? How is the recorded file saved and accessed on the device?

Q20. Signing, Versioning & Publishing What are the steps to publish an Android application on the Google Play Store? Explain: (a) why and how an APK/AAB is signed with a keystore, (b) the significance of versionCode and versionName in the manifest, and (c) the process of setting up a Google Play Console account, configuring the app listing, and submitting the app for review.